Nicolò Girardello

MSc student in Control Systems Engineering
Robotics & Embodied AI

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EDUCATION

Sep 2026 – Jul 2028 **MSc in Control Systems Engineering**
University of Padua, Italy
- Robot control, reinforcement learning, computer vision and deep learning for autonomous systems.

Sep 2023 – Jul 2026 **BSc in Automation and Systems Engineering**
University of Padua, Italy
- Control systems, robotics and machine learning, with strong foundations in mathematics, electronics and computer engineering.
- Thesis: *Tactile-to-Visual Representation Learning for Vision-Language-Action Models*.
- **Erasmus+** at UPC Barcelona (Sep 2025 – Jan 2026): mobile robotics, power electronics, applied MATLAB.

Sep 2018 – Jun 2023 **Technical Diploma in Electronics, Electrotechnics and Automation**
ITIS Alessandro Rossi, Vicenza, Italy

EXPERIENCE

Feb 2026 **Robotics Research Intern**
RoboTac Lab at BMW Group, Munich, Germany
- Worked on ROS2-based robotic systems in a research environment focused on intelligent autonomous systems.
- Contributed to system integration, motion planning and experimental validation of robotics applications.

Nov 2023 – Present **Technical Office Specialist**
MEP Solution Srl, Vicenza, Italy
- Designed and documented electrical and photovoltaic systems; technical analysis, cost estimation and cross-functional coordination.

SELECTED PROJECTS

Jul 2026 **Tactile-to-Visual Representation Learning for VLA Models** ↗
BSc Thesis · PyTorch, OpenVLA-OFT, DIGIT tactile sensors, MuJoCo, robosuite
- Multimodal framework that maps tactile signals into the frozen visual latent space of a VLA model, so touch can stand in for occluded or degraded vision.
- Built the simulation setup integrating tactile rendering in MuJoCo/robosuite on the LIBERO benchmark.

Jun 2026 **Redundancy Resolution for the Franka Emika Panda** ↗
Introduction to Robotics · ROS 2, C++, MATLAB, MuJoCo
- Kinematic control with CLIK and null-space projection to optimize arm posture without disturbing end-effector tracking.
- Validated in MATLAB and ported to a real-time C++ ROS 2 controller plugin.

Dec 2025 **Multi-Robot Warehouse Optimization Simulator** ↗
UPC Barcelona · MATLAB, App Designer
- Simulator with A*/Dijkstra path planning, priority-based collision avoidance, task assignment and battery-aware routing.

May 2022 **BerryBot: Automated Strawberry Harvesting** ↗
Siemens Automation Olympics · Siemens PLC, ABB arm, Python, OPC UA
- Computer-vision-guided robotic arm for greenhouse harvesting, integrated with a PLC and a remote web HMI.

ChessBot, a PLC-driven 3-axis chess robot, and more at girardellon.github.io/projects ↗

TECHNICAL SKILLS

AI & ML	PyTorch, Scikit-learn, computer vision, multimodal / contrastive representation learning, Vision-Language-Action models
Robotics	ROS2, MuJoCo, Gazebo, KDL, motion planning, kinematic control, multi-robot coordination
Programming	Python (NumPy, SciPy, Pandas), C / C++, MATLAB / Simulink
Automation	Siemens PLC (TIA Portal), OPC UA, PROFINET, LabVIEW
Languages	Italian, English, Spanish